

Nathaniel Kessel

Cincinnati, OH

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EDUCATION

University of Cincinnati

May 2028

Bachelor of Science, Computer Engineering

Miami Valley Career Technology Center (MVCTC), Clayton, OH

August 2021 – June 2023

- Completed FAA certified training for general, airframe, and power-plant aircraft technician licenses.
- Constructed a foundation for strict, reliable, and traceable documentation via logbooks and IAW references to FAA certified materials such as Type Certificate Data Sheets (TCDS) and manufacturer provided manuals

PROJECTS

STM32F4 3D Graphics Engine

September 2026 – Ongoing

Personal Project, [GitHub](#)

- Building a 3D graphics rendering engine based on the STM32F4 in bare-metal C and ARM Cortex M4 Assembly
- Initialized STM32 CMSIS and DPAL libraries, verified with a blink program before full graphics development
- Creating drivers for interfacing with the X-NUCLEO-GFX02Z1's ST7789V LCD controller from scratch
- Abstracting low-level hardware control with custom libraries mimicking ST's hardware abstraction layer (HAL)

ESP32-S3 Network Clock

September 2025 – September 2026

Personal Project, [GitHub](#)

- Engineered and iteratively debugged four hardware revisions of a custom printed circuit board (PCB) using KiCad, utilizing digital multimeters and oscilloscopes to verify signal integrity on SPI and I2C lines, power delivery, and trace routing
- Hand-soldered surface-mount (SMD) components, successfully integrating a custom SPI-driven LED display, onboard real-time clock (RTC), and NTP network time synchronization
- Developed firmware in Arduino C++, implementing NTP time sync, OpenWeatherMap API, and standard clock functions

Messaging Application, *Team Leader*

May 2026 – August 2026

University of Cincinnati, [Project Website](#)

- Architected a messaging web application utilizing Scrum and agile development practices
- Designed the database schema and real-time communication pipeline using Node.js, Socket.io, and MongoDB Atlas
- Established the planned deployment infrastructure via Azure App Services and GitHub Actions

EXPERIENCE

Closing Expert, **Target**, Milford, OH

September 2023- Present

- Balanced full-time academic course load with working part time at an average of 26 hours per week
- Trained and guided employees on WMS store metrics and how to utilize them for EOD performance

SKILLS

Languages: C, C++, Assembly (PIC18, ARM), Python, MATLAB, VBA, LabVIEW

Embedded Tooling: MPLABX, STM32CubeMX, ESP-IDF, Proteus, GNU Make, Bash

Hardware: KiCad 10.0, LTSpice, Multisim, Fusion360, AutoCAD, Soldering, 3D Printing

Collaboration and Version Control: Git, GitHub, GitHub Projects (Kanban + Scrum)

AVAILABLE TO CO-OP SPRING 2027 (JAN – MAY) & SUMMER 2027 (MAY - AUG)